

# PRATIK MANDANI

## HYBRID / PRODUCT DEVELOPMENT ENGINEER

Electronics + Embedded Firmware + Web Backend

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Portfolio: [pratik-mandani.github.io](https://pratik-mandani.github.io) • GitHub: [github.com/pratik-mandani/pratik-mandani](https://github.com/pratik-mandani/pratik-mandani)

### PROFESSIONAL SUMMARY

Hybrid / Product Development Engineer with 7 years of hands-on experience across electronics, vending-machine product development, hardware integration, machine testing, R&D and web application development. Started with hands-on production and electronics work, progressed into product development and R&D, and later expanded into PHP/Laravel full-stack development. Currently strengthening embedded firmware capabilities through personal ESP32 and C/C++ projects, including KOTL — a smart embedded assistant robot.

### CORE PROFILE

#### PRODUCT DEVELOPMENT

Vending-machine product development, machine integration, testing, wiring and R&D.

#### EMBEDDED

ESP32, C/C++, GPIO, ADC, I2C, SPI, UART and I2S.

#### ELECTRONICS

PCB prototyping, troubleshooting, repair, SMD and through-hole soldering.

#### WEB BACKEND

PHP, Laravel, MySQL, REST APIs, CRUD and business applications.

### TECHNICAL SKILLS

#### PRODUCT DEVELOPMENT & ELECTRONICS

Product Development, Vending Machine Systems, Machine Assembly, Machine Wiring, Hardware Integration, Hardware Testing, Product Testing, PCB Prototyping, PCB Design, PCB Repair & Troubleshooting, Component-Level Troubleshooting, SMD Soldering, Through-Hole Soldering, Electronics, Proteus 8

#### WEB & BACKEND

PHP, Laravel, Blade, MySQL, REST API, CRUD, Authentication, Middleware, Sessions, Admin Panels, E-commerce, Business / ERP Applications, HTML, CSS, JavaScript, Bootstrap

#### EMBEDDED & FIRMWARE

C, C++, ESP32, Arduino, GPIO, ADC, I2C, SPI, UART, I2S, Sensor Interfaces, Embedded Firmware Development

#### TOOLS & DEPLOYMENT

Git, VS Code, FTP, Hostinger, Proteus 8

### PROFESSIONAL EXPERIENCE

#### Altius Infoway — Rajkot, Gujarat

May 2022 – Present (4 Years)

##### Career Progression: Frontend Intern → Frontend Development → PHP/Laravel Development → Laravel Full-Stack Developer

- Developed and contributed to 100+ websites and web applications across corporate, e-commerce and business domains.
- Built backend applications using PHP and Laravel.
- Worked with MySQL databases and database-driven applications.
- Developed and integrated REST APIs.
- Worked on business management and ERP-style applications.
- Worked with clients on requirements, changes and issue resolution.
- Used Git and FTP in development/deployment workflows.
- Developed responsive frontend interfaces using HTML, CSS, JavaScript and Bootstrap.
- Developed CRUD modules and admin panels.
- Worked with authentication, middleware and sessions.
- Worked on e-commerce functionality.
- Performed bug fixing, maintenance and feature development.
- Managed website deployment, hosting and maintenance.

#### Symbian Vending Solutions — Rajkot, Gujarat

Nov 2018 – Mar 2022 (3 Years)

##### Career Progression: Assembly Worker → Production Team Leader → R&D Team Leader

Role: Product Development / R&D / Electronics / Machine Integration • Team: Approx 15 members

- Worked on vending-machine assembly, wiring, testing and operational validation.
- Coordinated procurement, dispatch and production requirements.
- Worked on PCB prototyping and electronics troubleshooting.
- Performed PCB repair and component-level troubleshooting.
- Coordinated with embedded, PLC, Raspberry Pi and application/web development teams during product integration.
- Reported product, production and technical issues to management.
- Coordinated day-to-day production activities with a team of approximately 15 members.
- Provided technical support for machine-related issues.
- Performed SMD and through-hole soldering.
- Worked on vending-machine product integration and R&D.
- Identified machine-level issues during testing and supported debugging and resolution.

## PRODUCT DEVELOPMENT PROJECTS — SYMBIAN VENDING

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### Reusable Vending Machine Control Card

PCB Prototype | Tool: Proteus 8

A small PCB prototype designed to make the vending control system adaptable across machines using common components and configurable connectors.

**Responsibilities:** PCB prototyping, connector design, hardware integration, testing and troubleshooting.

### 24V–12V Coin Mechanism Interface & Isolation Module

PCB / Power & Interface Module | Tool: Proteus 8

A custom interface module developed to integrate a 12V coin mechanism into a 24V vending-machine system.

**Design:** LM2576 power section for 24V → 12V conversion, Relay for 12V switching/control, Optocoupler for signal isolation between the coin mechanism and vending controller.

**Tool:** Proteus 8

### Touch / QR / Automatic Cup Vending Systems

Product Development / Machine Integration

Worked on vending-machine product development involving machine assembly, wiring, hardware integration, operational testing and coordination with embedded, application, PLC, Raspberry Pi and web development teams.

**Responsibilities:** Machine assembly, wiring, hardware integration, operational testing, developer coordination, R&D support.

## PERSONAL EMBEDDED PROJECT

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### KOTL — The Smart Emotional Assistant Robot

Active Personal Project

A personal embedded hardware and firmware project built to strengthen practical MCU, display, audio and sensor development skills.

**Hardware:** ESP32, SSD1306 OLED, MAX9814 Microphone, MAX98357A Audio Amplifier, Speaker.

**Firmware / Interfaces:** C/C++, GPIO, ADC, I2C, I2S.

**Current Implementation:** OLED RoboEyes-style interface, blinking, idle movement, sound detection, microphone input, audio playback and non-blocking audio playback.

## SELECTED WEB PROJECTS — ALTIUS INFOWAY

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**1. Diamorivo:** [Laravel + MySQL](#) — E-commerce platform. Frontend UI implemented from Figma design. Frontend + Backend.

**2. The Busa:** [Laravel + MySQL](#) — Fashion e-commerce platform. Frontend + Backend. Product CRUD, Cart, Order Management.

**3. Network Kitchenware:** [Laravel + MySQL](#) — Business management application. Dashboard, User CRUD, Authentication, Database integration.

**4. Luxuria Baths:** [PHP / Laravel / MySQL](#) — Sanitaryware catalog and business website. Frontend + Backend development.

Complete portfolio of 16+ featured platforms available at: <https://pratik-mandani.github.io/>

## EDUCATION

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### Diploma in Electronics & Communication Engineering

Passout: 2018

Gujarat Technological University (GTU)

## LANGUAGES & STRENGTHS

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**Languages:** Gujarati, Hindi, English

**Core Strengths:** Product Problem Solving, Hardware Troubleshooting, Cross-functional Coordination, System Integration, Debugging, Practical Engineering, Production Support, Client Communication, Technical Issue Resolution, Learning & Prototyping.